

MOUNTIAN EMPIRE, VA, USA

WMO: 724056

Lat: 36.895N

Lon: 81.350W

Elev: 2559

StdP: 13.39

Time Zone: -5.00 (NAE)

Period: 95-19

WBAN: 63805

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.8	13.4	-3.0	5.1	17.0	1.0	6.4	17.7	31.3	31.0	28.4	29.1	11.7	270	0.454

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	89.3	70.8	86.3	70.1	83.9	69.3	74.0	82.5	72.7	81.2	71.4	80.2	5.8	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.8	129.4	77.3	70.3	122.7	76.3	68.9	116.7	75.3	39.3	82.2	38.1	81.6	37.0	80.8	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.7	22.1	18.9	DB	2.9	90.7	6.9	3.7	-2.1	93.4	-6.1	95.5	-10.0	97.6	-15.0	100.3	(4)
			WB	1.8	76.3	6.5	1.9	-2.9	77.6	-6.7	78.8	-10.3	79.9	-15.1	81.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.3	33.6	37.0	45.0	54.1	62.5	69.7	73.1	72.0	65.8	55.3	44.1	38.0
	DBStd	16.15	11.81	10.62	10.38	8.86	6.97	4.91	4.36	4.53	6.30	8.73	9.00	9.73
	HDD50	1843	522	380	221	60	6	0	0	0	1	48	217	388
	HDD65	4721	974	785	622	337	128	18	4	5	66	317	627	838
	CDD50	3401	13	15	65	182	394	592	717	682	474	211	41	15
	CDD65	803	0	0	1	8	52	160	256	222	89	15	0	0
	CDH74	6336	0	1	18	118	479	1199	1908	1716	766	125	6	0
	CDH80	1720	0	0	1	7	72	305	579	537	201	17	1	0
Wind	WSAvg	6.9	10.2	9.6	8.8	8.1	5.9	5.0	4.8	3.8	4.2	6.0	7.4	8.8
Precipitation	PrecAvg	42.9	3.5	2.7	4.2	3.2	4.0	4.3	4.5	4.0	3.4	2.7	2.9	3.4
	PrecMax	56.1	8.1	5.4	7.9	5.8	7.8	8.4	7.8	7.2	12.2	9.1	6.2	7.4
	PrecMin	33.3	1.3	0.7	1.2	0.6	1.3	1.5	1.8	1.1	0.3	0.0	0.6	1.5
	PrecStd	6.2	1.8	1.2	1.6	1.3	1.9	1.9	1.4	1.7	2.4	2.2	1.6	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.1	70.1	76.7	81.1	85.5	90.1	91.1	91.2	90.1	82.1	72.8	66.8
		MCWB	57.6	55.9	58.9	61.8	68.1	70.3	72.3	71.6	67.6	65.6	58.3	57.3
	2%	DB	61.2	63.1	71.7	77.5	81.9	86.1	88.3	88.5	85.1	77.5	66.5	61.8
		MCWB	52.8	51.1	57.1	60.4	65.4	69.6	71.3	70.8	68.3	63.3	55.3	54.5
	5%	DB	55.4	57.4	65.9	73.7	79.4	83.6	85.7	85.8	81.8	73.2	63.6	57.2
		MCWB	50.0	49.0	53.5	58.5	64.4	68.6	71.3	70.0	66.5	60.5	53.7	51.3
	10%	DB	51.8	53.7	62.3	70.4	76.5	81.2	83.0	82.3	79.0	70.1	60.2	53.9
		MCWB	46.8	46.8	51.6	57.0	63.5	67.4	69.9	68.9	65.8	60.3	52.4	49.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.6	59.2	61.0	65.4	70.9	73.7	76.4	75.0	72.8	70.6	61.3	60.1
		MCDB	64.2	65.6	72.2	75.2	79.5	81.9	85.1	84.8	81.4	76.2	68.5	64.4
	2%	WB	55.0	55.4	58.8	63.0	68.7	72.1	74.6	73.4	70.6	66.9	58.9	56.2
		MCDB	59.9	60.9	67.9	73.5	77.6	81.1	82.7	82.7	78.8	73.5	64.6	60.3
	5%	WB	51.2	51.1	56.3	60.7	67.0	70.7	73.1	72.1	68.9	63.9	56.5	53.1
		MCDB	55.3	55.8	64.3	70.6	75.9	80.0	81.1	81.5	77.7	70.8	61.9	56.3
	10%	WB	46.8	47.3	53.7	58.8	65.2	69.2	71.8	70.8	67.3	61.4	52.9	49.0
		MCDB	50.5	51.9	60.3	67.7	73.8	78.1	80.0	80.0	76.3	67.8	58.6	52.6
Mean Daily Temperature Range	5% DB	MDBR	17.0	17.7	21.1	22.5	21.8	20.3	19.1	20.4	21.4	22.0	20.9	16.9
		MCDBR	22.2	25.6	27.6	28.0	25.9	24.4	22.5	24.8	26.6	27.4	27.3	23.2
		MCWBR	16.1	17.0	15.9	14.6	13.1	11.1	10.0	11.1	12.2	14.7	17.0	16.2
	5% WB	MCDBR	19.6	20.9	23.3	22.6	21.1	20.0	18.7	20.2	21.7	20.0	20.7	18.8
		MCWBR	15.9	16.6	14.7	12.7	11.2	9.9	9.0	9.6	10.3	11.7	15.2	15.1
Clear-Sky Solar Irradiance	taub	0.290	0.297	0.321	0.364	0.404	0.433	0.464	0.469	0.367	0.334	0.301	0.283	
	taud	2.521	2.488	2.468	2.363	2.250	2.197	2.142	2.103	2.386	2.481	2.544	2.564	
	Ebn at Noon	289	300	300	291	279	270	260	256	281	282	282	285	
	Edh at Noon	25	29	33	38	43	46	48	49	35	29	24	23	
All-Sky Solar Radiation	RadAvg	703	906	1258	1600	1762	1911	1779	1656	1412	1108	829	613	
	RadStd	60	110	83	126	135	131	119	84	132	135	82	59	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (42 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+248	+148

Nomenclature: See separate page